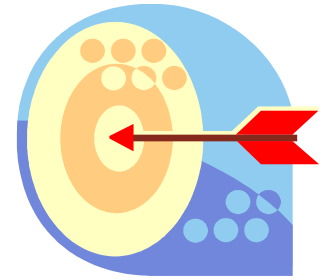


Lesson 2 Basic BPML Concepts

Introduction:

This lesson provides an overview of the components that make up the Business Process Modeling Language (BPML)



Objectives:

- ✓ Define BPML
- ✓ Discuss the activities that perform BPML functions
- ✓ Differentiate between services and adapters as they relate to Sterling
 B2BIntegrator
- ✓ Explain how Sterling B2B Integrator uses BPML

What is BPML

BPML is short for Business Process Modeling Language. It is an XML-based language for describing a business process. You will need to know the basics of XML. For example, you'll need to know about elements and attributes, usage of start, end, and empty tags, and other basic XML concepts. Advanced features such as DTD structure or various entity references are not needed.

An understanding of XPath is necessary to access document content and to build rules. For more information about XPath, see Lesson 3, XPath Basics.

What is BPML – Services and Adapters

Services and adapters are software components that can be used to accomplish integration or business tasks. Business processes make calls to services and adapters to accomplish your business tasks. For a business process to interact with these components, you must include services as steps in your business process.

A service is a component of Sterling B2B Integrator, such as a translator. Services that access disk or other applications (reach outside of Sterling B2B Integrator) are known as **adapters**.

Process Data Vs. Primary Document

Process Data

- Acts as a virtual memory for the business process.
- Is always in XML
- Information can be **assigned** to it or retrieved from it to be used as variables in your business process.

Primary Document

- Document being worked on. Such as an EDI doc to be translated.
- Can be any type of data
- Is an object link in Process data.
- A service/adaptor will always work on Primary Document.

Message Exchange



Message Exchange

- BPML engages a message-based business process in which everything that interacts with a process does so through the exchange of messages. In Sterling B2B Integrator, business processes interact with services/adapters and other business processes.

Exchanging messages with servers/adapters

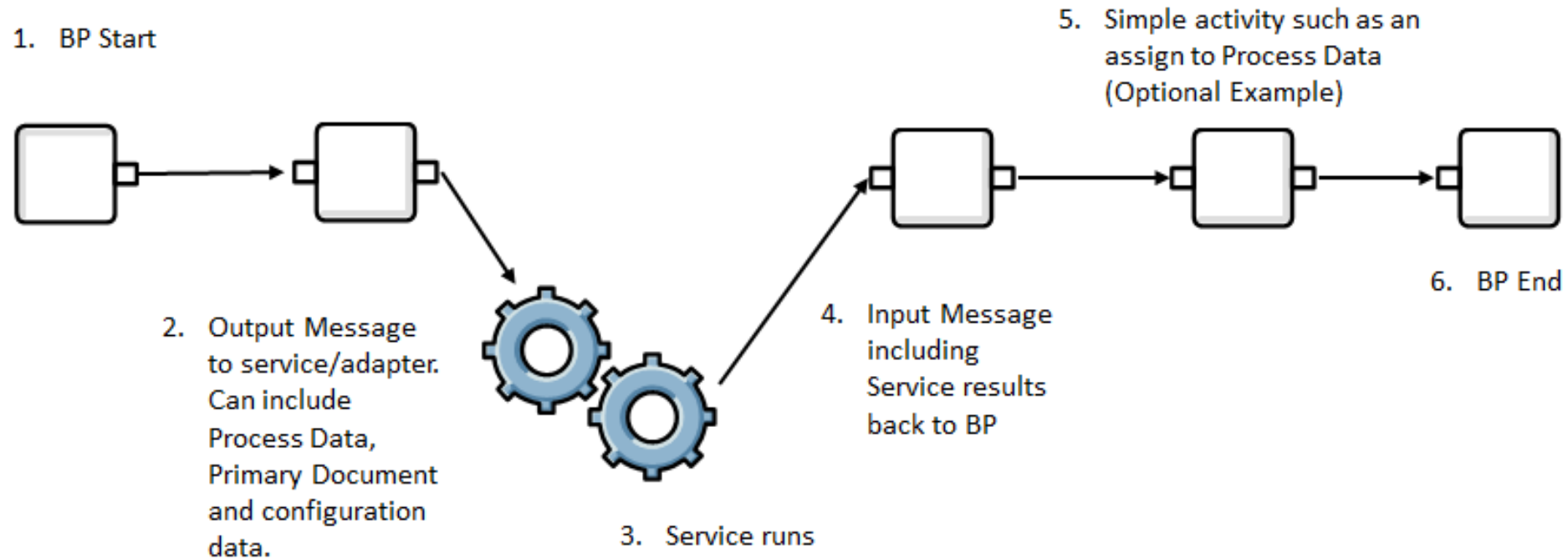
- In Sterling B2B Integrator, whenever a business process runs an adapter or service, there is an exchange of two messages.
- Business Process Context to server/adapter
 - Configuration data
 - Overrides
 - Primary Document
- Server/adapter back to Business Process Context
 - Results from service executing
 - Updated/same Primary Document

Exchanging messages with other business processes

- Messages are exchanged with business processes by way of the produce and consume BPML activities.
 - The produce activity sends a message
 - The consume activity receives a message from a process and consumes it into the child process.
- While BPML supports this message exchange, it is a best practice in Sterling B2B Integrator to use the **Invoke Business Process service** to start a child or parallel process and pass message information.

Message Exchange Example

Business Process Flow



Sterling B2B Integrator BPML Activities

Overview:

Activities perform BPML functions. BPML functions include operations that allow for the control of the flow of data through a business process. For example, the choice activity allows data to flow through the system based on the result of a decision.

BPML supports three main types of activities:

- ✓ Simple
- ✓ Complex
- ✓ Process

Simple Activities

Use simple activities such as produce, consume, and assign to perform only one function. For example, the produce activity only sends a message to another process. The assign activity is used only for setting a value.

Example:

```
<assign to="X">7</assign>
```

This assign would set the value of “7” to a tag <X> in Process Data.

```
<X>7</X>
```

Complex Activities

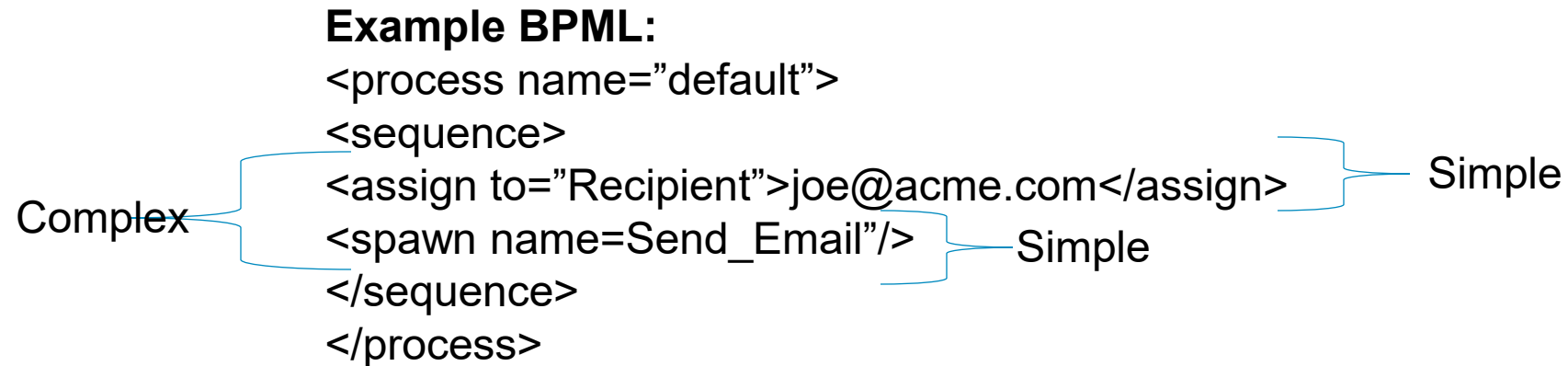
Use complex activities, such as sequence, all, and choice to model the flow of control. Complex activities can consist of one or more simple activities or another complex activity. These types of activities can be sequential, parallel, or conditional.

Example BPML:

```
<process name="default">  
  <sequence>  
    <assign to="Recipient">joe@acme.com</assign>  
    <spawn name=Send_Email"/>  
  </sequence>  
</process>
```

Sequence Activity

A sequence activity runs all of the activities contained in it when a process is run. The activities are run in the order they are listed in the sequence.

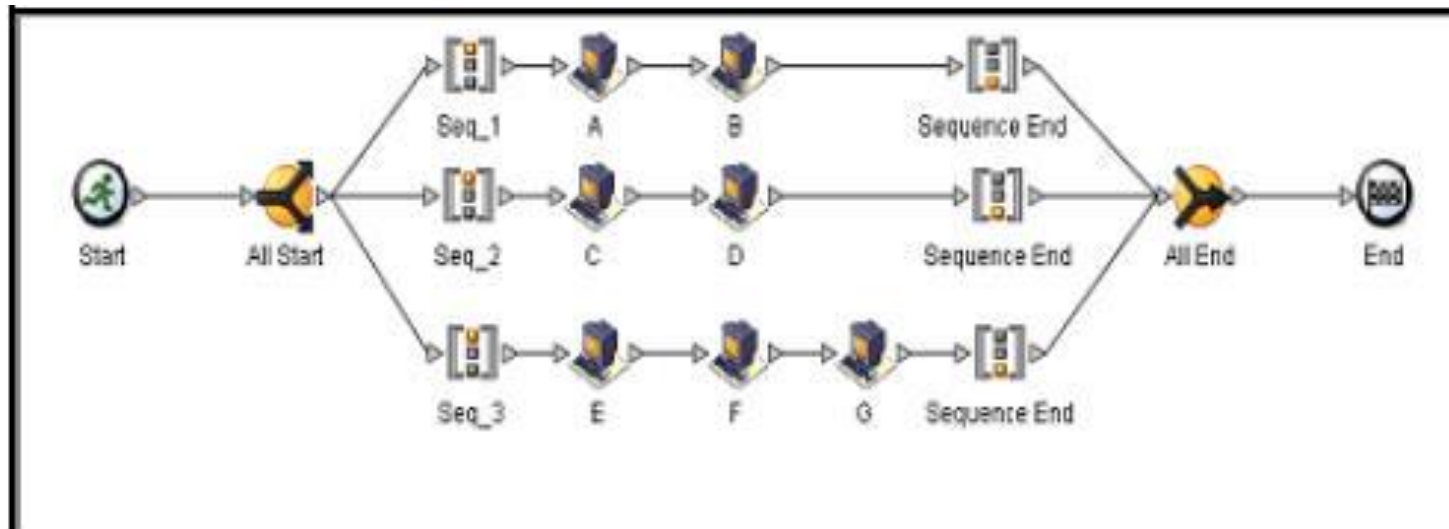


In this case we have a sequence that has two activities, Assign and Send E-mail. When the process is run, the system first runs Assign then runs Send E-mail because that is the order the activities are listed in the sequence.

BPML All Activities

The All activity is a complex activity that does not complete until all its sub, or child activities are run. In Sterling B2B Integrator, All runs all of its child activities simultaneously.

- Use the All activity any place where an activity can appear.
- You must specify two or more sub activities.



BPML All Activities

(continued)

Example BPML:

```
<all>
  <sequence name="Subflow1_Start">
    <Subflow1>.....</Subflow1>
  </sequence>
  <sequence name="Subflow2_Start">
    <Subflow2>.....</Subflow2>
  </sequence>
  <sequence name="Subflow3_Start">
    <Subflow3>.....</Subflow3>
  </sequence>
</all>
```

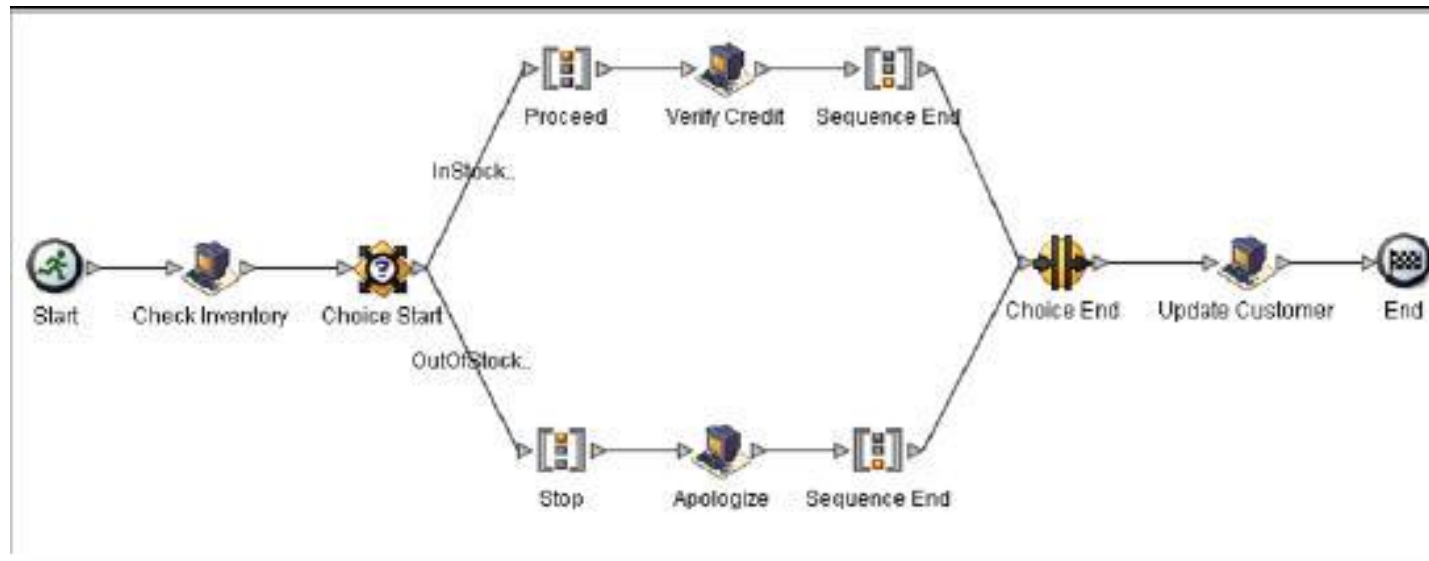
Using the BPML All Activity Properly

- The BPML activity, All (represented in the GPM by the All Start and All End icons), runs two or more complex child activities within a business process simultaneously.
- Because there are performance implications to using the All activity, use it only if you expect branches of your business process model to take significant amounts of time (in seconds or minutes) to complete (very few services take this long).
- Do not use the All activity to run two or more instances of the same subprocess simultaneously. The system will handle the proper load balancing.

BPML Choice Activities

The Choice activity is one of three types of complex activities. (The others are Sequence and All) A Choice activity is used to make decisions in the business process model and runs only one of the child activities it contains. The Choice activity makes it possible to model branch processing.

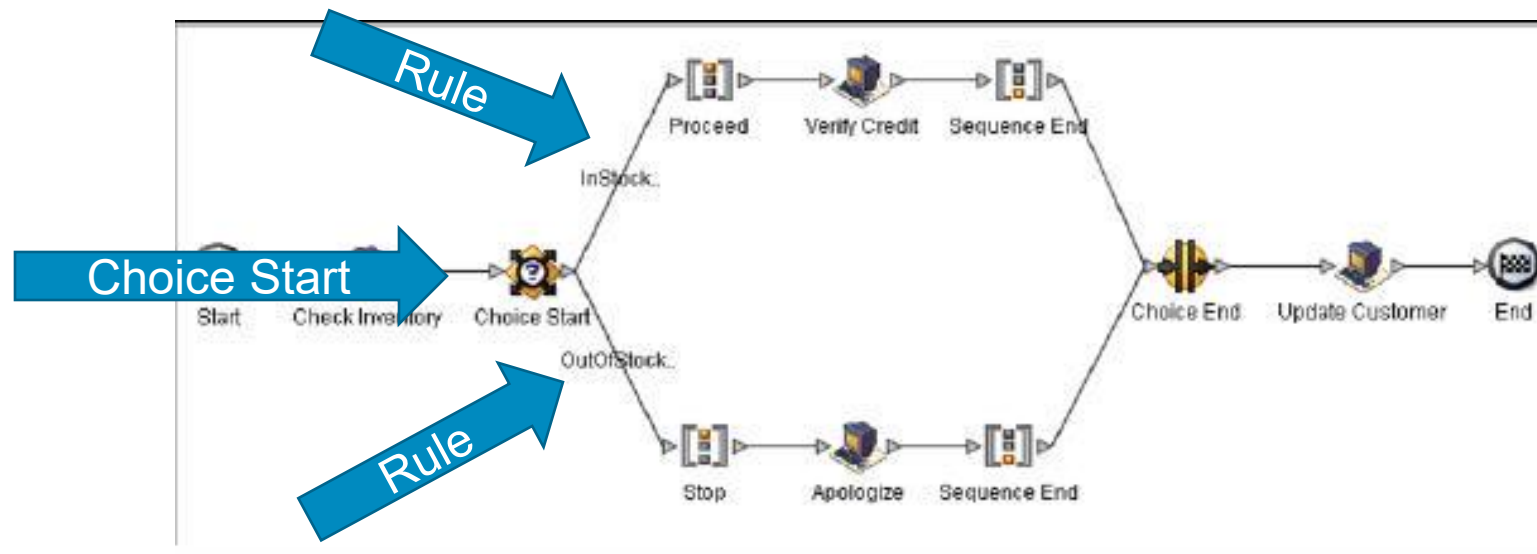
- Use the Choice activity any place where an activity can appear.
- Use a choice to model process branching.



BPML Choice Activities

(continued)

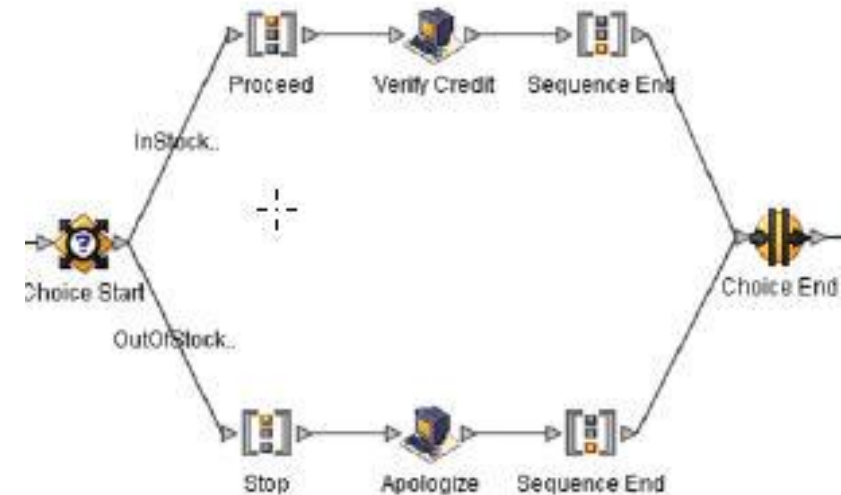
Choice Start means that one or more rules are needed so that a decision can be made. Can be referred to as an If, Then statement.



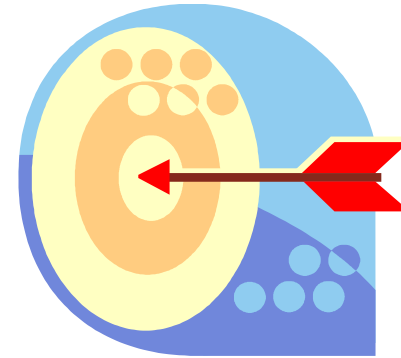
BPML Choice Activities (continued)

Multiple cases can reference the same activity. However, the activity will run exactly once. If an activity is not referenced, it will never run. If it is determined that no activity is candidate for running, the choice completes immediately. In Sterling B2B Integrator the order in which the case elements appear is the order that they are run. Rules are necessary with choice elements and determine the business process decision making.

```
<choice>
  <select>
    <case ref="BookInStock" activity="proceed"/>
    <case ref="BookInStock"
      negative="true" activity="stop"/>
  </select>
  <sequence name="proceed">
    <operation name='Verify Credit Card'>
    </operation>
  </sequence>
  <sequence name="stop">
    <operation name='Apologize'> ... </operation>
  </sequence>
</choice>
```



Activities Supported by BPML



Sterling B2B Integrator recognizes several different BPML activities. Please pause audio and review the descriptions in your student manual for the references provided.

Process Rules

A rule is an expression that evaluates single or multiple conditions that are based on information known to the process or information communicated to the process. The conditional expressions allow for diverse business process decisions both internal to the process and external to the environment.

Rules express single or multiple conditions that are based on information known to the process or information communicated to the process. The contents of the condition activity must correspond to an expression that can be evaluated as either true or false. BPML mandates that these expressions be XPath expressions. Complex activities often require that a process select one of several alternate activities or even, on occasion, discriminate the information upon which it acts. Rules can effect activity selection using the choice activity.

Process Data

How is Process Data used and what is it's purpose:

Process data is a context that exists for each process instance and can be used to hold or reference information that is accumulated during the life of the process. Process data is a private area that the process definition writer uses to store data that can be accessed by the running business process.

Data entering the system can be of many types, including a document and/or a set of name/value pairs. For example, when you run a business process from the User Interface, you can select an input file to use as a document. When a business process is started, it has data it puts into the document, and it can also set name/value pairs in the process data.

Services often use Process Data to store those name/value pairs that services down the line in the business might need to use as variable data.

Using the Assign Element

- As a process runs, it obtains information from participants and communicates it to other participants. In Sterling B2B Integrator, participants can be services, adapters, or other processes.
- An incoming document may contain a great deal of data, but the process participants may only need a small portion of that data. Therefore, the process needs to be selective about what it actually shares with different participants.
- The process decides what information is relevant and uses the Assign element to share that subset. This takes a known value and includes it in an output message. In a similar manner, it can take a value from an input message and make it available for the process.
- The direction in which an assignment is done depends on the context in which the Assign element appears; if it is part of the input, the assignment is from message to process; if it is part of the output, the assignment is from process to message.



Using the Assign Activity

- The Assign activity sets something in the process data equal to a fixed value, either a number or a string; additionally, it can incorporate XPath expressions. When using the assign element in an operation, you must know the parameters for the service. An example of this is the name of a map that the translator service needs.
 - An operation is usually a call to a service/adaptor and would represent a configuration setting.
 - The assignment source and target contexts depend on the manner in which the assign element is used.
 - When the assign element appears within the output element, it performs assignment from the process data to the outgoing message. The To attribute corresponds to the path within the message and is used to construct the message contents.
 - When the assign element appears within the input element, it performs assignments from the incoming message to the process data. The To attribute corresponds to a value within the process data.
 - When the assign activity appears as an activity, it performs assignment from a constant to the process data. The To attribute corresponds to a path within the process data. The From attribute can be used to extract information from a previously assigned value.
 - Sterling B2B Integrator does not recognize nested assigns.

Using the Assign Activity - Example

Example:

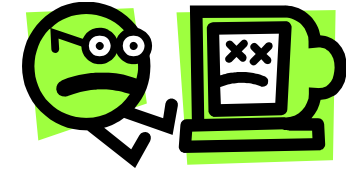
The schema for the assign activity as an independent activity is:

```
<assign to = "x"> 3 </assign>
```

The schema for the assign element within an operation is:

```
<operation name="name">  
  <participant name="name of specific service"/>  
  <output message-"output message from service"/>  
  <input message="input message for service">  
    <assign to='z' from ="x"/>  
  </input>  
</operation>
```

Fault Handling



- A fault is an error that occurs while running any step of a process regardless of whether it is the result of a local error or an error in a service/adaptor. Fault handling allows the process to recover and proceed. A process may simply communicate the failure to its participants, or it could engage in recovery or select a different path to run. Generally, our main goal in setting up a fault handling procedure is to track any error and inform the operator.
- Faults are propagated upwards through a process from the faulting activity to its parent activity. If the parent activity does not have an appropriate onFault to handle the error, it is propagated up to its parent, and so forth. If several activities are running in parallel, only the activity with the fault is affected. However, when the top-level activity fails, the process is aborted. Any activity associated with the process will complete, but the process will still abort.

Using the OnFault Activity

- Errors are handled in BPML by the onFault activity. An onFault activity is associated with the complex activity it is contained in, and is only run if an activity in the complex activity results in an error.
 - The fault handling activity runs if the activity faults before completing. The faulting activity completes if the fault handling activity completes successfully.
 - Use multiple onFault activities to handle fault codes. A set of onFault activities must use unique fault codes and at most one activity may omit the code attribute.
 - The onFault activity without a code runs only if no fault with a matching code is found in the correct complex activity.

Example:

```
<onFault code?>  
activity+  
</onFault>
```

Process Rules

(continued)



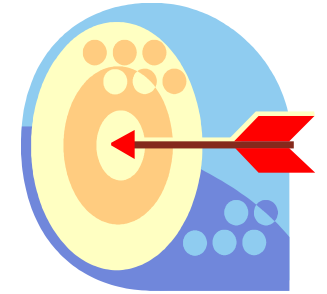
Exceptions:

An exception indicates a possible failure condition. A service may need to generate a workflow exception when a required input parameter is:

- Missing.
- Invalid.
- Disabled. For example, a map or a workflow definition is disabled.

Example: There is a PO from Joes Shirts requesting several shirts in red, white and blue. The company filling the order has all the red and all of the white but only some of the blue. The PO should not be stopped because they don't have all the blue.

Element Editors



BPML activities are installed and configured automatically when you install the GPM. However, some BPML activities, services, and service types may require you to perform additional configuration in the GPM, depending on the task your process model needs to accomplish. You can locate these configuration parameters in one of five different element editor types: **Property, Service, Edge, Produce, and Consume**. Each element editor provides configuration parameters specific to an activity or a service.

Knowledge Check

1. How many branches within a Choice can be executed as a result of a case statement?
 - a. One
 - b. Two
 - c. Three
 - d. Unlimited
2. Services and adapters are used to exchange messages between business processes.
 - a. True
 - b. False
3. The Produce activity does not wait for the message to be consumed.
 - a. True
 - b. False
4. The All activity does not complete until all the sequences are complete.
 - a. True
 - b. False

Knowledge Check

5. Which of these elements are available in Operations of services/adapters?

Select all the correct answers.

- a. Output
- b. Participant
- c. Input
- d. Message To Service
- e. Message From Service

6. "UseXpath" should be selected whenever you add some parameter to the Release services.

- a. True
- b. False

7. The parent and child processes in a Spawn activity are independent of each other.

- a. True
- b. False

Lesson Summary



In this lesson we:

- Learned the definition of BPML
- Became familiar with the actives that perform BPML functions
- How to differentiate between Services and Adapters as they relate to Sterling B2B Integrator
- Can explain how Sterling B2B Integrator uses BPML

**Congratulations, you have completed Lesson 3,
Basic BPML Concepts!**